Assignment 8 – Git Basics

Task 1: Initialize a Git Repository

The `git init` command initializes a new Git repository in the project directory, allowing Git to track changes to files.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git --version
git version 2.52.0.windows.1
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git init
Initialized empty Git repository in C:/Users/Aryan/OneDrive/Documents/assign8-gi
t/.git/
PS C:\Users\Aryan\OneDrive\Documents\assign8-git>
```

Task 2: Configure Git

The `git config` command sets the user's name and email address, which are recorded in commits.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git config user.name "SARAIYA ARYAN MIRAL"
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git config user.email "2025ucp1712@mnit.ac.in"
```

Task 3: Create and Stage a File

A `readme.txt` file was created to describe the project. The `git add` command stages the file, and `git status` displays the current state of the repository.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git add readme.txt
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   readme.txt
```

Task 4: Commit Changes

The `git commit` command saves the staged changes as a commit in the repository's history. The `git status` command confirms that the working directory is clean.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git commit -m "Adding first file."
[master (root-commit) 9533da2] Adding first file.
 1 file changed, 2 insertions(+)
 create mode 100644 readme.txt
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git status
On branch master
nothing to commit, working tree clean
PS C:\Users\Aryan\OneDrive\Documents\assign8-git>
```

Task 5: View File Changes

After modifying `readme.txt`, the `git diff` command displays the differences between the working file and the staged version, helping identify the changes made.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git diff
diff --git a/readme.txt b/readme.txt
index 5f1be4a..4b68193 100644
--- a/readme.txt
+++ b/readme.txt
@@ -1,2 +1,3 @@
 Vector Dot Product Project
-This project calculates the dot product of two vectors using Python.
\ No newline at end of file
+This project calculates the dot product of two vectors using Python.
+Git tracks changes made to project files.
\ No newline at end of file
```

Task 6: Explore Git Restore

The `git restore` command discards unstaged modifications and restores a file to its last staged state. The `git restore --staged` command removes changes from the staging area without discarding the modifications in the file.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git restore readme.txt
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git add readme.txt
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git add readme.txt
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git add readme.txt
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   readme.txt

PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git restore --staged readme.txt
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   readme.txt

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git add readme.txt
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git commit -m "Update project description"
```

```
[master e5baa53] Update project description
 1 file changed, 2 insertions(+), 1 deletion(-)
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git status
On branch master
nothing to commit, working tree clean
```

Task 7: View Commit History

The `git log` command displays the repository's commit history, including commit hashes, authors, dates, and commit messages.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git log
commit e5baa5339f7b326a7daf244c7165067776c5b81f (HEAD -> master)
Author: SARAIYA ARYAN MIRAL <2025ucp1712@mnit.ac.in>
Date:   Wed Oct 28 14:28:40 2026 +0530

    Update project description

commit 9533da2efcadc5fb00b30fd68b35620ebff8317c
Author: SARAIYA ARYAN MIRAL <2025ucp1712@mnit.ac.in>
Date:   Wed Oct 28 13:59:24 2026 +0530

    Adding first file.
```

Task 8: Revert a Commit

The `git revert` command reverses the changes introduced by a selected commit by creating a new commit. The original commit remains in the repository history.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git revert e5baa5339f7b326a7daf244c7165067776c5b81f
[master 81d612d] Revert "Update project description"
 1 file changed, 1 insertion(+), 2 deletions(-)
```

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git log
commit 81d612db05f16ef7b562df24cf72235474d0ef40 (HEAD -> master)
Author: SARAIYA ARYAN MIRAL <2025ucp1712@mnit.ac.in>
Date:   Wed Oct 28 14:34:00 2026 +0530

    Revert "Update project description"

    This reverts commit e5baa5339f7b326a7daf244c7165067776c5b81f.

commit e5baa5339f7b326a7daf244c7165067776c5b81f
Author: SARAIYA ARYAN MIRAL <2025ucp1712@mnit.ac.in>
Date:   Wed Oct 28 14:28:40 2026 +0530

    Update project description

commit 9533da2efcadc5fb00b30fd68b35620ebff8317c
Author: SARAIYA ARYAN MIRAL <2025ucp1712@mnit.ac.in>
Date:   Wed Oct 28 13:59:24 2026 +0530

    Adding first file.
```

Task 9: Commit the Python Program

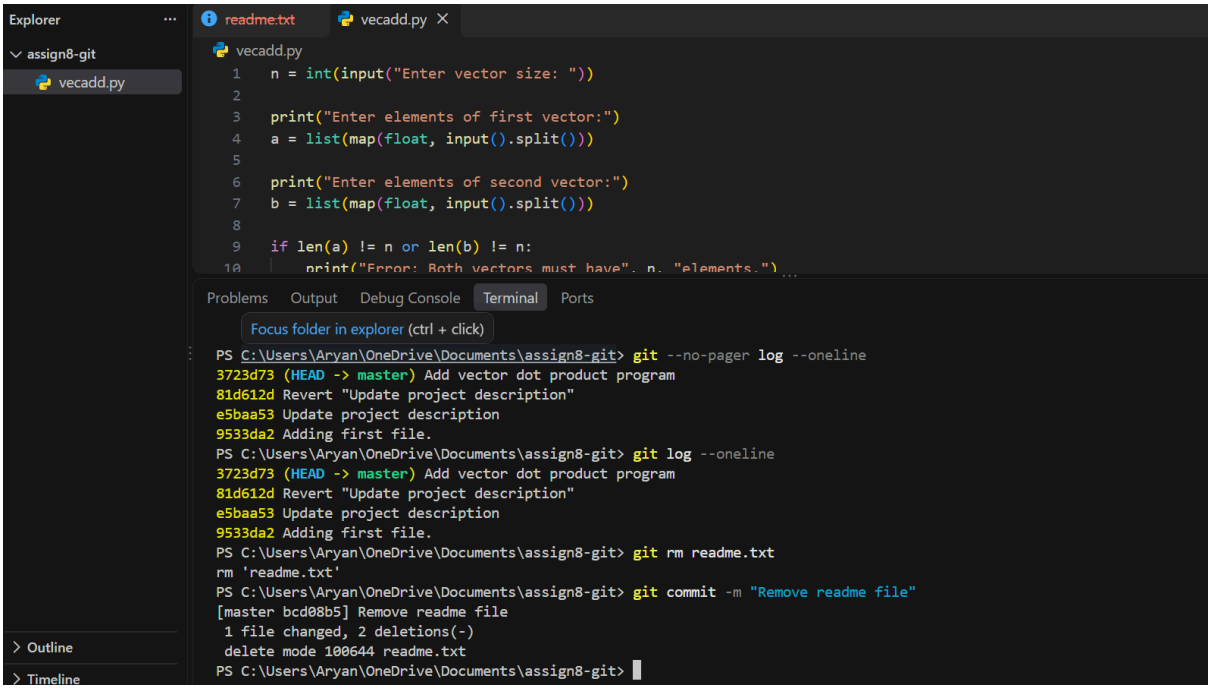
The `vecadd.py` file was created to calculate the dot product of two vectors. The file was staged and committed using Git, and the commit history was displayed to verify the changes.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> python vecadd.py
Enter vector size: 2
Enter elements of first vector:
2
Enter elements of second vector:
3
Error: Both vectors must have 2 elements.
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> python vecadd.py
Enter vector size: 2
Enter elements of first vector:
1 2
Enter elements of second vector:
2 3
Dot product: 8.0
```

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git log --oneline
3723d73 (HEAD -> master) Add vector dot product program
81d612d Revert "Update project description"
e5baa53 Update project description
9533da2 Adding first file.
```

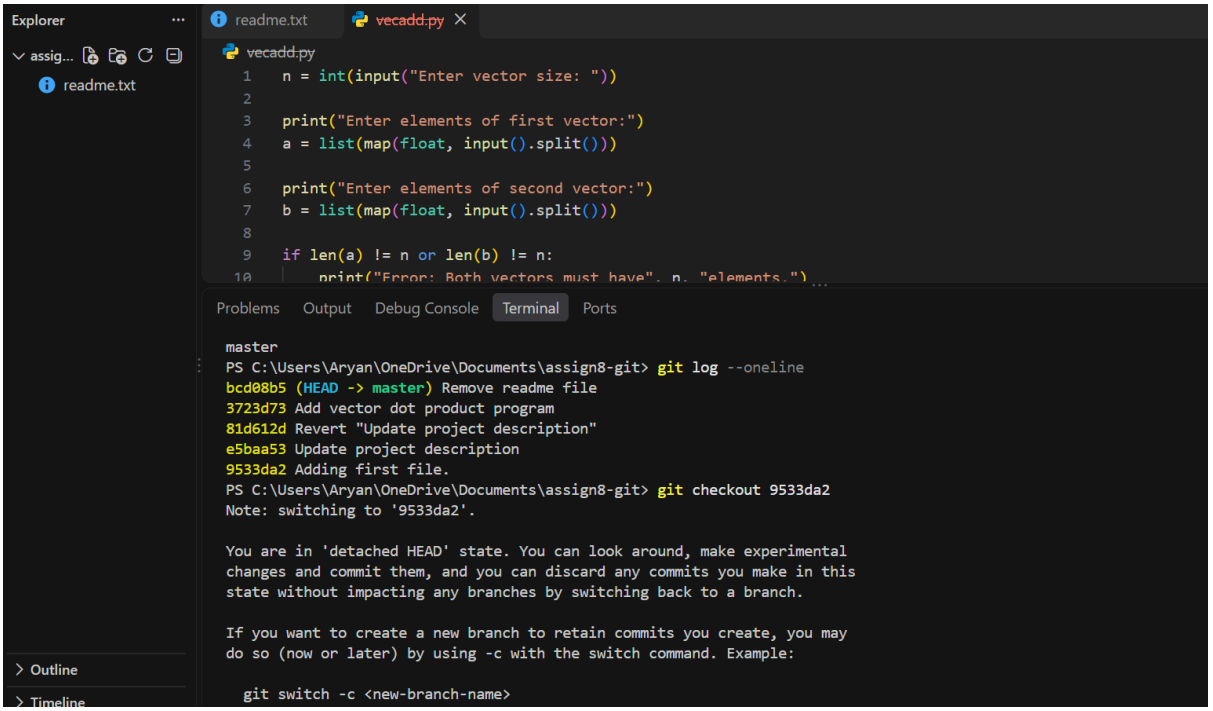
Task 10: Remove a File

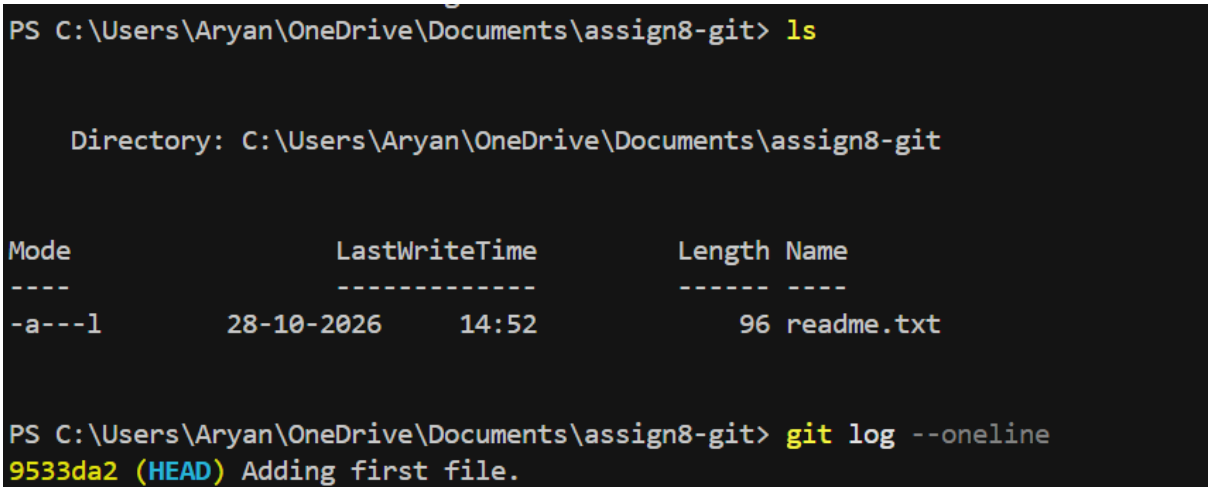
The `git rm` command removes `readme.txt` from the working directory and stages its deletion. A new commit records the removal in the repository history.



Task 11: Check Out an Earlier Commit

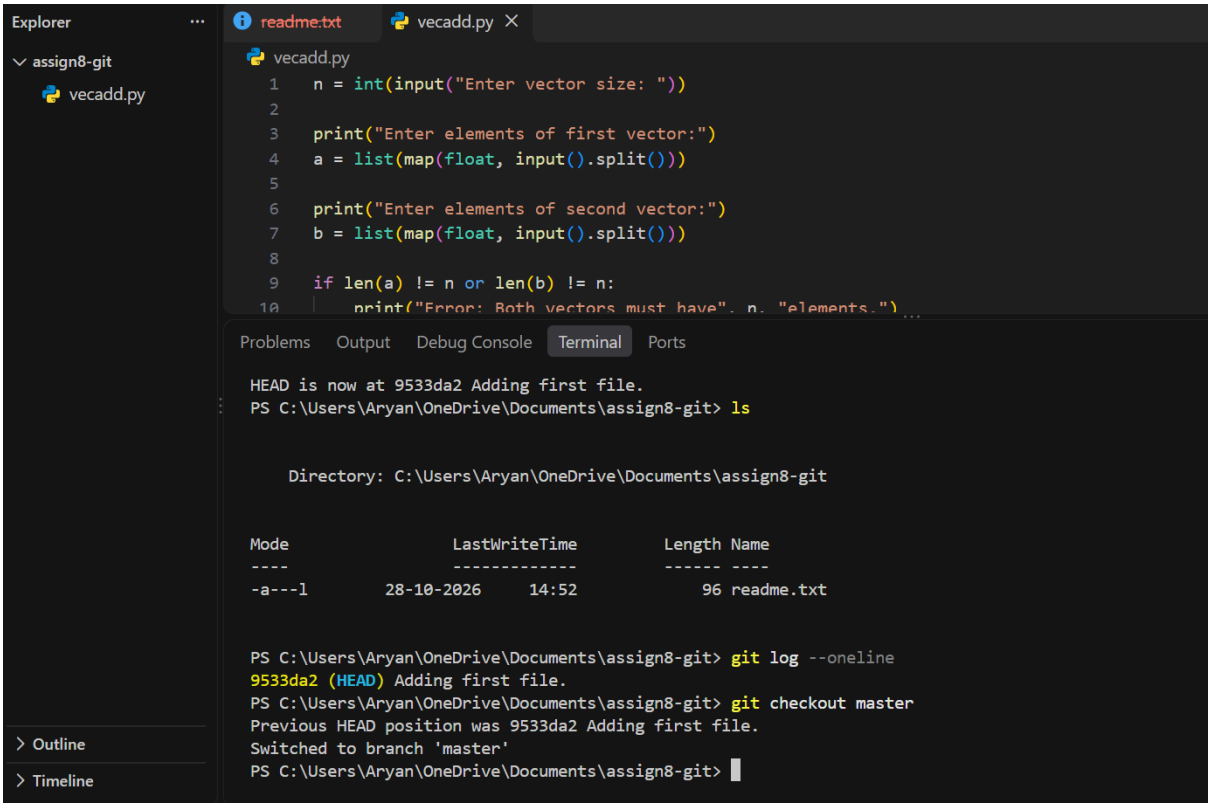
The `git checkout` command was used with an earlier commit hash to view a previous version of the project. The files and commit history were examined to observe the state of the repository at that point.





Task 12: Return to the Latest Commit

The `git checkout` command was used with the branch name to return to the branch's latest commit. The repository status and commit history were checked again.



Conclusion

This assignment provided practical experience with basic Git commands for repository initialization, configuration, staging, committing, comparing changes, restoring files, viewing history, reverting commits, deleting files, and checking out earlier versions of a project.